

Certified Reference Material CERTIFICATE

IAEA-603

Calcite

Certified values of IAEA-603

Parameter	Certified value	Combined standard uncertainty ($k=1$)
$\delta(^{13}\text{C})_{\text{VPDB}}^{(a)} [\text{‰}]$	+2.46	0.01
$\delta(^{18}\text{O})_{\text{VPDB}}^{(a)} [\text{‰}]$	-2.37	0.04

^(a) The expression of the quantity carbon and oxygen isotope delta follows Skrzypek et al.[1] Alternative expressions for the quantity, e.g. for oxygen, include $\delta(^{18}\text{O}/^{16}\text{O})$ [1] and more commonly $\delta^{18}\text{O}$ following published guidelines [2].

Origin and preparation of the material

A slab of Carrara marble, composed predominantly of the mineral calcite, was obtained by the IAEA in 1990 from Italy. This material was initially used in the production of the reference material IAEA-C1 and subsequently for IAEA-CO-1.

For the preparation of IAEA-603, approximately 80 kg of the remaining coarsely crushed raw marble were further processed. The material was ground and sieved to obtain approximately 15 kg of a grain-size fraction between 200 μm and 500 μm . The selected fraction was thoroughly homogenized and subdivided into 31 portions.

Each portion was carefully washed with deionised water to remove fine particles, oven-dried, and stored in 15 glass flasks. For the preparation of the first batch of IAEA-603 (approximately 2.8 kg), the contents of three flasks were recombined, homogenized, and dispensed into glass ampoules, each containing approximately 0.5 g of calcite. The ampoules were flushed with argon to minimize atmospheric contamination and were subsequently flame-sealed. This first batch comprises approximately 5,000 ampoules. The remaining material (approximately 12 kg) was transferred into four flame-sealed glass containers for long-term storage and future use.

Note 1: The prepared grain-size fraction (200–500 μm) consists of clumped smaller calcite crystals, reflecting the natural structure of the material. The grain may be disaggregated by vigorous shaking without affecting the suitability of the material.

Feasibility Study

A feasibility study was conducted at GEOTOP–UQAM (Montreal, Canada) on the initially sieved material. One sample from each of the 31 prepared portions was selected and shipped to GEOTOP prior to the final re-mixing and ampouling at the IAEA. At GEOTOP, the material was further disaggregated and re-sieved into grain-size fractions. Four sub-fractions (below 90 μm , 90–125 μm , above 125 μm , and bulk) were analysed using approximately 120 μg aliquots. The samples were reacted with H_3PO_4 (density 1.93 g/cm³) at 90 °C. The results demonstrated that IAEA-603 is homogeneous at this sample size within the observed standard deviations (0.02 ‰ for $\delta(^{13}\text{C})_{\text{VPDB}}$ and 0.04 ‰ for $\delta(^{18}\text{O})_{\text{VPDB}}$; $n = 31$ for each size fraction).[3] It was therefore concluded that the material is suitable for use as an international reference material, including applications involving small sample sizes down to 120 μg within these uncertainty limits.

Homogeneity of the material

A heterogeneity study of the final ampoules (first batch of 5,000 ampoules) was subsequently carried out at the IAEA. In accordance with ISO 33405 [4] 52 ampoules were randomly selected from the batch of 5,000 and at least two independent analyses were performed per ampoule. Twenty aliquots (10–12 mg each) were analysed in parallel by reaction with 2.5 mL H_3PO_4 (density 1.93 g/cm³) at (25 ± 0.1) °C for 24 hours, as described into details in Assonov et al 2020. [3]

This heterogeneity study yielded standard deviations of ± 0.009 ‰ for $\delta(^{13}\text{C})_{\text{VPDB}}$ ($n = 195$) and ± 0.035 ‰ for $\delta(^{18}\text{O})_{\text{VPDB}}$ ($n = 148$), based on both the total dataset and individual analytical sequences (20 parallel aliquots).

It was concluded that:

- the reported standard deviations incorporate both analytical uncertainty and any uncertainty arising from potential heterogeneity at the tested aliquot size (10–12 mg);
- the material can be considered homogeneous within the observed variability, with any contribution from heterogeneity being equal to or smaller than the measured standard deviations.

Characterization study

The characterization of IAEA-603 relative to NBS 19 was carried out by the IAEA laboratory on the final product (IAEA-603 sealed in ampoules). The measurements were performed under the same conditions (10–12 mg sample size) as those used in the homogeneity study. [3]

For confirmation of assigned values and validation purposes, results from three additional expert stable isotope laboratories were considered:

GEOTOP-UQAM, Montreal, Canada –feasibility study performed on material sampled prior to final re-mixing and ampouling, using a sample size of 120 µg

USGS, Reston, USA – study conducted on material sampled before final re-mixing and ampoules, with a sample size of 20 mg;

MPI-BGC, Jena, Germany – study performed on five ampoules selected from the final product, using a sample size of 40 mg.

Stability of the material

Short- and long-term stability was not experimentally assessed based on the proven stability of similar materials, like NBS 19. This component is assumed to be negligible, as the material is inherently stable and storage in sealed ampoules prevents exchange with atmospheric moisture and CO₂ [3].

Assignment of values

The assigned values were calculated from the results obtained by a reference measurement procedure (DI-IRMS) in one laboratory (IAEA) in accordance with ISO 33405 [4].

Table 2 summarise the assigned values and the contributors to the assigned uncertainty:

Table 2. Results of the IAEA-603 characterization at the IAEA.

	$\delta(^{13}\text{C})$ of IAEA-603, ‰ on VPDB scale ⁽¹⁾	N ₁ / N ₂ - number of measurements on IAEA-603 and on NBS19	$\delta(^{18}\text{O})$ of IAEA-603, ‰ on VPDB scale ⁽¹⁾	N ₁ / N ₂ - number of measurements on IAEA-603 and on NBS19
Characterization study (95%-confidence intervals)	2.460±0.005	N ₁ =38 (10-12 mg) N ₂ =38 (10-12 mg)	-2.373±0.017	N ₁ =38 (10-12 mg) N ₂ =38 (10-12 mg)
Heterogeneity study (1 standard deviation)	±0.009	N ₁ = 195	±0.035	N ₁ = 148
Assigned values, including the material heterogeneity (1k uncertainties)	2.46±0.01		-2.37±0.04	

The Reference Materials (RM) Certification Committee accepted in 2016 these assigned values as certified values as presented in this certificate.

The combined uncertainty of the assigned values includes the following components:

- The uncertainty associated with characterization relative to NBS19 (conservatively estimated at the 95% confidence level);
- The uncertainty derived from the heterogeneity study (see above);
- The analytical uncertainty, which was not separated from potential inhomogeneity. Instead, the standard deviations observed during the heterogeneity study were taken as a combined contribution of inhomogeneity and analytical uncertainty;

Results from GEOTOP-UQAM, USGS, and MPI-BGC were in agreement, within uncertainties, with the assigned values (Table 2) [3].

Statement on metrological traceability and uncertainty of assigned values

The $\delta(^{13}\text{C})$ and $\delta(^{18}\text{O})$ values and associated uncertainty are traceable to the VPDB delta scale, defined by taking the fixed numerical value of +0.00195 for $\delta(^{13}\text{C})$ in the NBS 19 Reference Material, and by taking the fixed numerical value of -0.0022, for $\delta(^{18}\text{O})$ in the NBS 19 Reference Material in reactions with oversaturated phosphoric acid at 25 °C [5].

The measurement uncertainty associated with the assigned values is expressed as a combined standard uncertainty (coverage factor $k = 1$) estimated in accordance with JCGM 100:2008 [6] and ISO 33405 [4].

Absolute isotopic abundances

The absolute isotopic abundances have not been determined.

Intended use

The reference material IAEA-603 was developed to replace the quarantined reference material NBS 19. Its intended uses are as follows:

- To serve as the highest-level metrological reference material for the practical realization of the VPDB and VPDB-LSVEC $\delta(^{13}\text{C})$ scales and VPDB and VPDB-CO2 $\delta(^{18}\text{O})$ scales [5], with the purpose to calibrate analytical systems at end-user laboratories;
- To serve as the highest-level reference material for the characterization of lower-level reference materials (secondary reference materials and internal laboratory standards) for $\delta(^{13}\text{C})_{\text{VPDB/VPDB-LSVEC}}$ and $\delta(^{18}\text{O})_{\text{VPDB/VPDB-CO2}}$ analysis.

Note:

Based on the results obtained from both unseparated material and sub-fractions sampled at 120 μg [3], IAEA-603 is also suitable for use as an international reference material at this sample size. This is particularly important because the material may gradually disintegrate into finer particles over time, for example as a result of handling or shaking.

Instructions for use, handling and storage

The reference material IAEA-603 is supplied in 0.5 g units sealed in flame-sealed glass ampoules. The ampoules should be stored at ambient temperature. Upon first use, end users are advised to transfer the material from the opened ampoule into the capped vial provided for subsequent storage.

Any glass fragments introduced during ampoule opening do not affect the $\delta(^{13}\text{C})$ and $\delta(^{18}\text{O})$ values of the CO_2 released from the material by acid digestion, nor the $\delta(^{13}\text{C})$ value of the CO_2 generated by thermal decomposition.

For the preparation of CO_2 gas used in the calibration of analytical systems, the material may be digested with H_3PO_4 or thermally decomposed using an elemental analyzer (EA) when calibration of $\delta(^{13}\text{C})$ only is required. The reaction time for the carbonate– H_3PO_4 digestion must be adjusted according to the analytical method and instrumentation employed to ensure complete reaction of IAEA-603. Under the conditions used at the IAEA (carbonate– H_3PO_4 reaction with phosphoric acid of density 1.93 g cm^{-3} at 25°C), IAEA-603 was found to react slightly faster than NBS 19.

After opening the ampoule and transferring the material into the storage vial, the material should be stored dry at ambient temperature. To minimize storage-related effects, such as isotopic exchange with atmospheric moisture and CO_2 , the vial must be kept in a desiccator. Maintaining the long-term isotopic integrity of the material after opening the ampoule is the responsibility of the user.

Limit of distribution

Only one unit may be ordered by one laboratory within a calendar year. This strategy helps ensure that the material remains available for worldwide use for as long as possible.

Issue and expiry date

The issue date of IAEA-603 is August 2016. The assigned values have been confirmed internally and by other 2 laboratories.[7] Based on this and experience with similar materials, the expiry date is set to July 2036, provided the original ampoules are stored undisturbed, in accordance with the instructions given in this certificate (see “Instructions for Use”). The IAEA is monitoring the long term stability of the material and customers will be informed in case of any observed change.

Legal disclaimer

The IAEA shall under no circumstances be liable for any loss, damage, liability or expense incurred or suffered that is claimed to have resulted from the use of information contained in this Reference Material Certificate.

Citation of this certificate

It is suggested to cite this certificate according to the following example, as appropriate to the citation format used: INTERNATIONAL ATOMIC ENERGY AGENCY, Certified Reference Material Certificate IAEA-603, IAEA, Vienna. (The latest version published applies, see “Note” below).

Note

Certified values as stated in this certificate may be updated if more information becomes available. Users of this material should ensure that the certificate in their possession is current. The current version is available on the IAEA's Reference Products website:

Further information

For further information regarding this material, please contact the producer:

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References

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Revision information

Revision 1: extension of expire date and update of the text to the current version of certificate template

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The project officer is responsible for the content of this certificate.
The Chair of the RM Certification Committee approves this certificate and authorizes its release on behalf of the Committee.